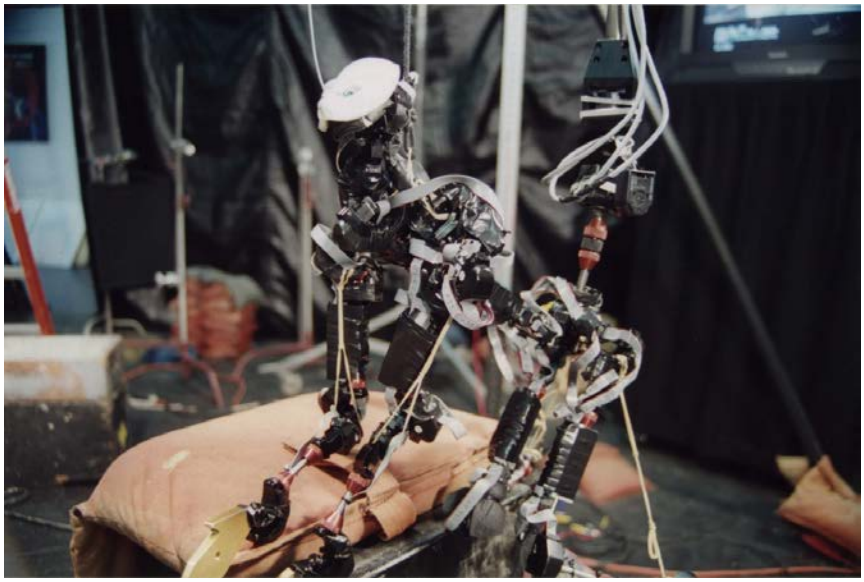
RETRO

When motion capture puppets were all the rage

July 13, 2020 · by Ian Failes



A look back at Boss Film's input device for Species' Sil.

You might remember *befores & afters'* coverage of the Dinosaur Input Device (DID) developed for *Jurassic Park* and also used by Tippett Studio on *Starship Troopers*. While the DID is probably one of the most well-known motion capture puppet input

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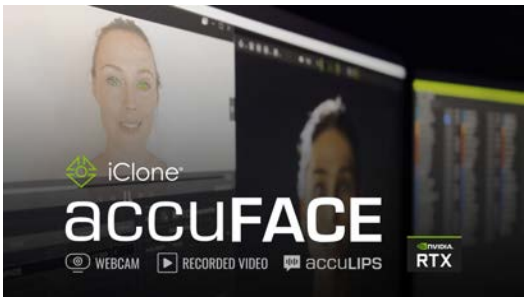

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devices, there were a few other ingenious armatures employed around that time, too, including a device made by Richard Edlund's Boss Film Studios for Roger Donaldson's *Species*. The movie is celebrating its 25th anniversary right now.

Boss Film made their motion capture device – a 2-foot high articulated puppet fitted with A/D potentiometers that was connected to an overhead armature and operated by four puppeteers – to animate scenes of the alien Sil creature. It was 'wired in' directly into Alias Wavefront to a proxy CG version of Sil and real-time composited against live-action backgrounds, complete with match-moved cameras, allowing for multiple takes which the director could review instantly.

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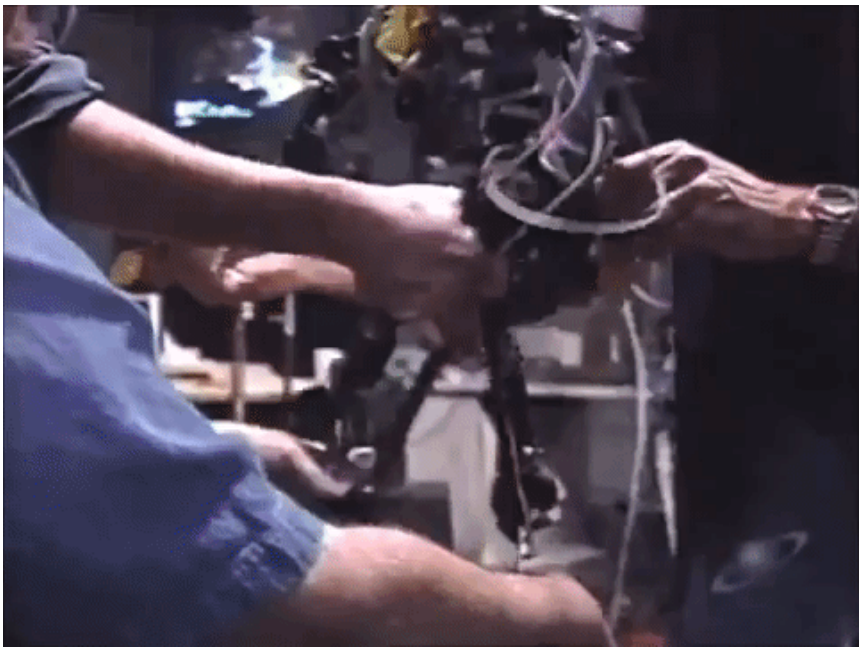

CG and puppetry meet

For Boss Film, which had for many years advanced the art of optical visual effects and then jumped head-first into the new world of CG, the motion capture puppet served as a cross between two eras of effects filmmaking.

"We had done a bunraku rod puppet for *Alien 3* shot under motion control," recalls André Bustanoby, who at Boss Film was a design engineer and eventual head of physical integration and responsible for the mechanical side of the Sil puppet. "Then with *Species* we were going to an input device, but being puppeteered in real-time, with some of the same puppeteers from *Alien 3*. So this was like a natural extension for us."

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André Bustanoby with a dinosaur armature used for a test shot for the film 'Dinosaur'. (Image courtesy André Bustanoby)

Bustanoby adds that Boss Film had also experimented with a motion capture puppet armature for a test shot for Disney's *Dinosaur* (the film would ultimately be released as a CG/live-action hybrid in 2000).

"We put together a quick armature with potentiometer sensors and then it had a spatial arm that had this dinosaur puppet on the end of it and it would move along a track. All this piped into Wavefront in real-time. It became this panning shot as the dinosaur walks by the camera through palm groves and such, kind of a darker version of what ILM had done as a test a couple of years earlier with *Jurassic Park*."

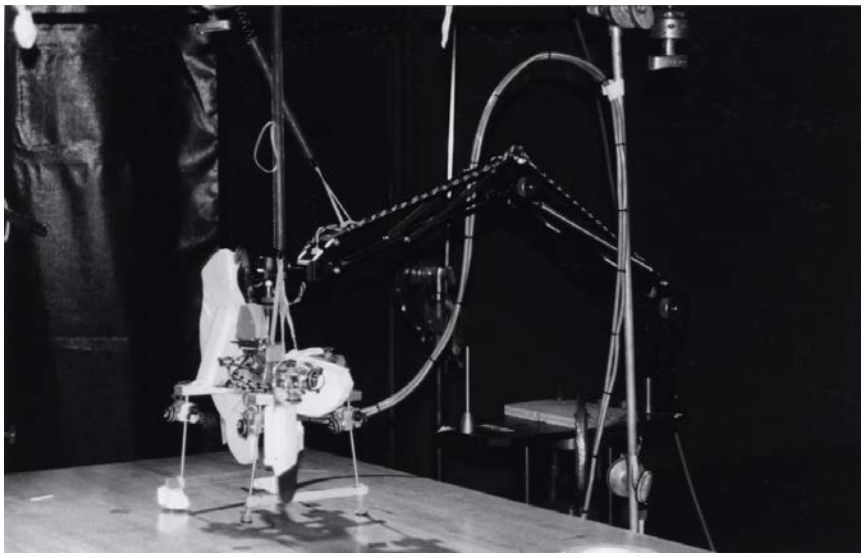
Boss Film's setup area for the 'Dinosaur' test. From left to right: Shahril Imbrahim (at the SGI), Mike Wise, Jeff Platt and Laine Liska. (Image courtesy André Bustanoby)

The CG dinosaur model into which the input device fed animation. (Image courtesy André Bustanoby)

That armature was later broken down for a breakfast cereal commercial that Boss Film worked on that required a CG cow creamer dish. When *Species* came around, the team at the visual effects studio took the performance side of the armature much further.

The goal, shared Bustanoby, was to be able to create multiple iterations of the performance and provide

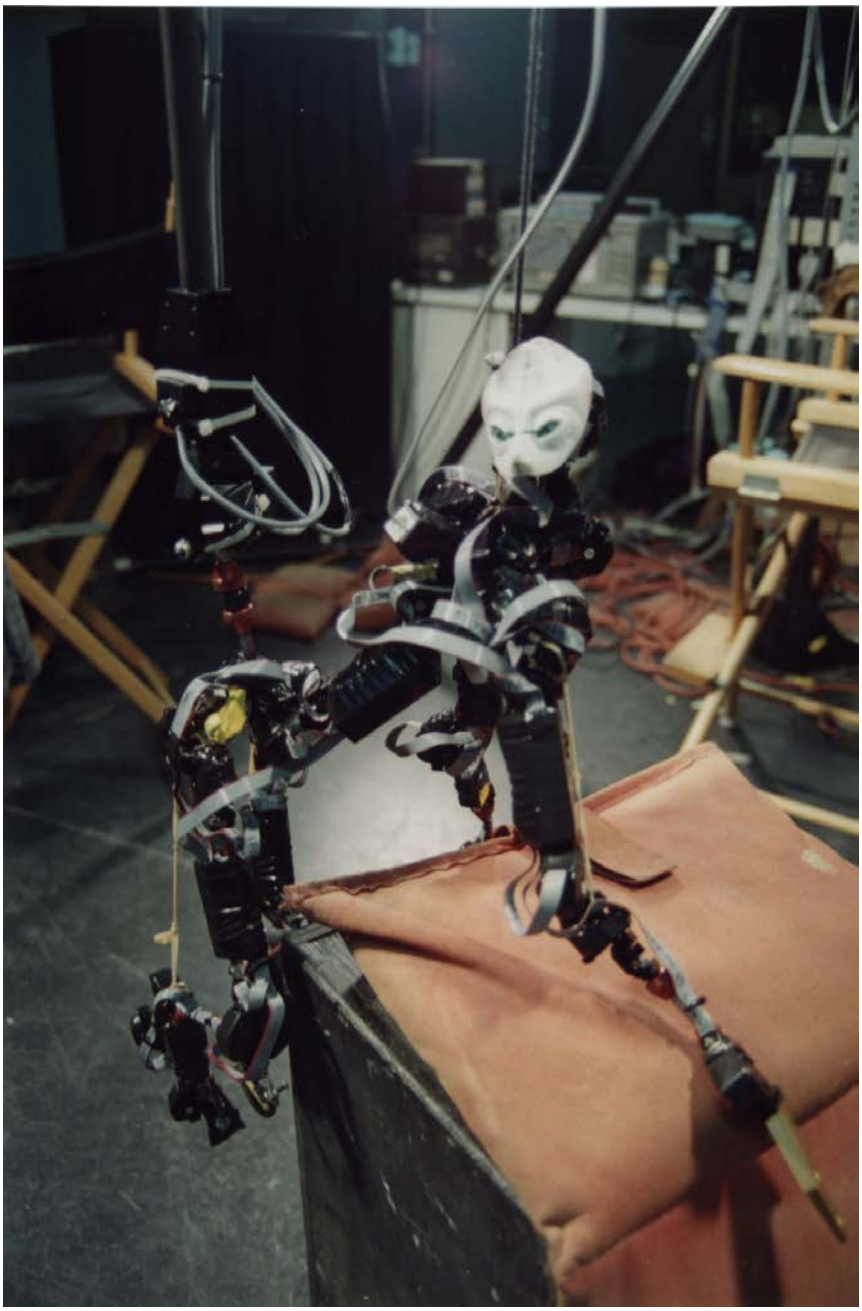
the director with instant feedback on the animation, rather than showing him animation dailies that would require lengthy turnaround times.



The input device used for the cow character in the breakfast cereal commercial. (Image courtesy André Bustanoby)

How the Sil puppetry worked

To provide that kind of fast feedback, a workflow was established that began with pre-tracked live action plates using early 3D tracking software. "This was then fed into the puppet rig, which was all in Wavefront," explains Bustanoby. "So we were doing not just real-time capture and performance, we were also doing real-time comps via a high-performance SGI mainframe complete with a Galileo card."



The Sil armature. (Image courtesy André Bustanoby)

Four puppeteers handled Sil, a puppet in the shape of the creature (it had been designed by H.R. Giger, with a full-scale version crafted by Steve Johnson's XFX). Two versions of the Boss Film armature were ultimately made, which differed in their types of bearings and overall mechanical layout. For Sil's joints, Bustanoby referenced the idea of 'Tinkertoys' and the fact that they were made up of lengths that could change, the idea being that the armature could

be broken down after the show and used for a different character with differing morphology and proportions.

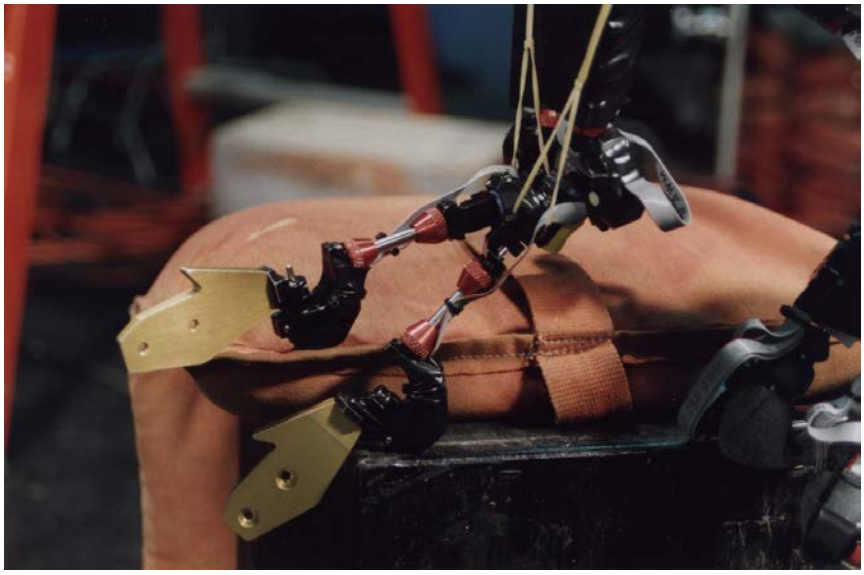
"Every joint axis, all 42 or however many there were – we pretty much covered everything except the fingers and individual vertebrae – was broken down into different areas," details Bustanoby. "We broke it down into cervical, thoracic, lumbar etc. We weren't going to cover all 40 something individual spine vertebrae. Then we had a 15 or 16 foot track that would carry this arm that held the armature at its hip. It was this long probe popping right into the hip area so that we could globally position the skeleton as it ran, jumped, walked or just stood there."

A view of the armature and its attachments to the computer. (Image courtesy André Bustanoby)

"Every joint had a little A/D brain, a little circuit board. It would literally read in real-time all the data coming off as it moved; there were three per joint, so X-Y-Z. It would send that off via a ribbon cable, back up along the arm and back to the SGI computer and feed into Wavefront."

"I will never forget the day that we hooked all this stuff up," continues Bustanoby, "including the main

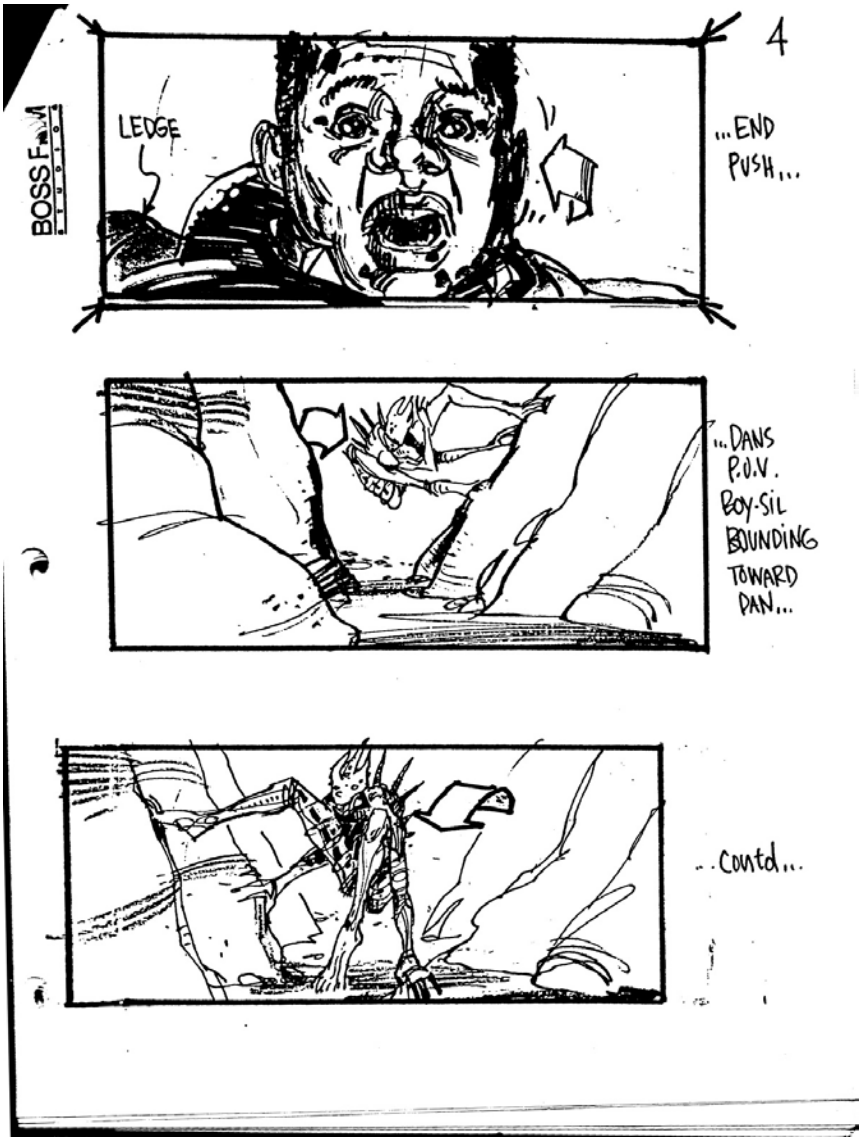
carriage, which had a big rotation drum and that went to an arm and another joint down another three feet and then another three feet. I held my fingers on the toes because I wanted to see, when I looked at Wavefront, if, when I moved the arm and all the joints, whether those feet at the very, very end of that kinematic chain would sit on the ground and be locked. And they did."



Close-up on the armature. (Image courtesy André Bustanoby)

The puppeteers would rehearse scenes and then keep iterating until they got a performance they were happy with. "It was a lovely thing to watch because it became a dance for them," observes Bustanoby. "It would be almost like they were performing music, counting out beats."

That motion captured data – along with data from a hand-operated Waldo rig used to provide lip and facial performance – provided Boss Film artists with the animation of the character. From here they lit, rendered and composited the final CG Sil into the live-action plates.



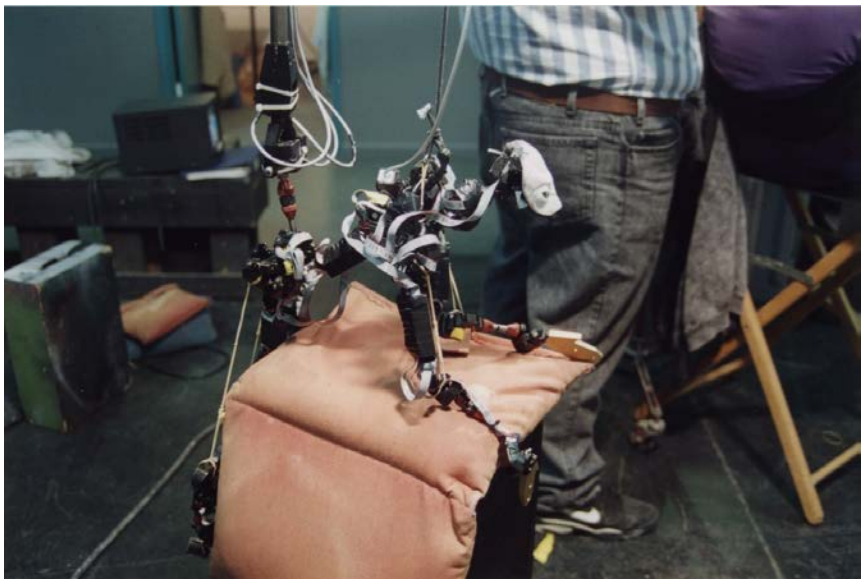
Boss Film 'Species' storyboard. (Image courtesy Trey Stokes)

The puppeteers operate the Sil armature. From left to right: Trey Stokes, Craig Talmy, Doug Miller and Bill Hedge. (Image courtesy André Bustanoby)

What became of Boss' digital puppetry

Bustanoby says that after Species the plan had been to utilize the armature – in a re-jigged fashion – for other creature work. This did occur for a spider character in a Boss Games video game (Boss Games was a Redmond-based spin-off company).

Unfortunately, it was announced in August 1997 that Boss Film Studios would close down arising from the incredibly competitive nature of the visual effects industry. As a result, most of the studio's assets were auctioned off to the public. "I remember going to the auction where the entire Sil rig sold," says Bustanoby. "I bid on it, actually, but it wasn't to be. I sometimes wonder where it went. Working on that was definitely a highlight for me."



The Sil puppet at the ready. (Image courtesy André Bustanoby)

Species 'Sil' – Performance / Armature / Systems Team

Motion Capture Puppeteers

Craig Talmy: Lead Motion Capture Puppeteer
Bill Hedge: Motion Capture Puppeteer
Doug Miller: Motion Capture Puppeteer
Trey Stokes: Motion Capture Puppeteer

Electronics/Engineering

Jeff Platt: Electronics Supervisor
Mike Wise: Electronics Engineer
Donny Sierer: Electronics Technician

Mechanical/Engineering

André Bustanoby: Head of Physical Integration / Design Engineer
Ken Dudderar: Engineering Foreman
Paul Smith: Engineering Precision Machinist

Software/Engineering

Shahril Ibrahim: Head of Software Development
Amit Agrawal: Software R&D
Chris Ross: Software R&D
Gautham Krishnamurti: Software R&D
Hiroyuki Miyoshi: Software R&D
Meher Gourjian: Software R&D

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